Venkata Lakshmi Parimala Pasupuleti

Full-Stack Software Engineer · AI Eval Infrastructure · Python · React · TypeScript · LangGraph · FastAPI · Observability · AWS · CI/CD

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PROFESSIONAL SUMMARY

Full-stack Software Engineer with 8+ years in production systems, specializing in AI evaluation infrastructure and agentic platforms. Built AgentProbe: open-source AI eval toolkit covering exactly the product surface HUD is building — browsing model behavior, inspecting trajectories, reviewing task quality, debugging failures, surfacing pipeline health — with a CI-integrated 5-metric eval engine, LLM-as-judge scoring, trace-level observability, and golden dataset regression harness. Built LeaseI: 8-agent LangGraph platform with durable state, tool orchestration, fault-tolerant recovery, and HITL escalation, evaluated at 93.6% precision across 1,240 test cases. At Entrata, shipped 20+ production modules full-stack (Python, JavaScript, React, PostgreSQL, AWS) across 120+ Agile sprints at 100K+ MAU. M.S. Data Analytics Engineering, George Mason University (GPA 3.87). — github.com/venkatalakshmiparimala/AgentProbe

WORK EXPERIENCE

Blue Horse Digital Sep 2026 – Present

AI/ML Engineer (Volunteer) · Remote

- Building ML models, evaluation pipelines, and full-stack APIs surfacing insights from raw business data into application-facing reporting layers.

Magizhs Tech Solution May 2025 – Aug 2025

AI/ML Engineer Intern

- Built agentic automation system (LangGraph + FastAPI) for real-world client — shipped to production, iterated weekly on customer feedback. Sole engineer, full ownership from day one.

Entrata (PropTech / Real Estate SaaS) Sep 2018 – Aug 2024

Senior Software Engineer · Remote

- **Full-stack production ownership across 20+ modules** (Python, PHP, JavaScript, React, PostgreSQL, RabbitMQ, AWS) serving 100K+ monthly active users across 120+ Agile sprints — designed, implemented, tested, deployed, operated, and debugged end-to-end without hand-holding. Shipped in environments where a wrong output means a compliance failure for thousands of residents.
- **Built and operated observability + debugging tooling** (OpenTelemetry, Prometheus, Grafana) for mission-critical data pipelines — cut mean time-to-diagnose by 40%; built debugging tools that reduced bug count to below 10 for 6+ consecutive months. Every production incident resolved to permanent root-cause fix.
- **Fault-tolerant event-driven systems** (RabbitMQ) with retry logic, idempotency, and state recovery across concurrent account updates; Docker microservices redesign (50% scalability improvement, 40% deployment time reduction); 200+ SQL optimizations (30% performance gain); 80% manual workload eliminated via Python/Bash automation.
- Mentored 5+ engineers; 300+ peer code reviews; built internal tooling that made the team dramatically faster.

Entrata India, Pune Jul 2017 – Sep 2018

Associate Software Engineer

- Built backend modules (PHP, Python, Java, PostgreSQL); led AWS S3 migration; data pipelines across hundreds of thousands of records.

PROJECTS

AgentProbe: AI Evaluation + Observability Infrastructure EVAL PLATFORM CI-INTEGRATED

Python · FastAPI · LangGraph · React · ChromaDB · GitHub Actions · OpenTelemetry | github.com/venkatalakshmiparimala/AgentProbe

- 15-endpoint FastAPI backend; 5-metric eval engine (correctness, groundedness, hallucination rate, latency, token efficiency); LLM-as-judge scoring with golden dataset annotation; CI regression harness with pass/fail gating; trace-level trajectory observability; distributed model routing across OpenAI/Anthropic/Groq/Ollama. Built from scratch because the tooling didn't exist — same high-agency instinct HUD describes.

Leasel: RL-Style Agent Evaluation — Trajectories, Grading + HITL LANGGRAPH RL-ADJACENT EVAL

Python · LangGraph · FastAPI · SQLite · RAG · OpenAI API | github.com/venkatalakshmiparimala/Lease_complaince_rag

- 8-agent StateGraph generating multi-step rollout trajectories (Triage → Retrieval → Configuration → Historical → Reasoning → Validation → Human escalation) with durable state across steps. Built a grader pipeline on top: LLM-as-judge scores each trajectory outcome against ground truth, flags low-reward rollouts for HITL review, and gates regressions via a pass@k harness (multi-trial sampling, 93.6% precision, 87.9% F1, pass@3 88.3% across 1,240 episodes). Every trajectory step is traced, scored, and auditable — the same data collection + grading + quality review loop HUD builds at scale.

EDUCATION

M.S. Data Analytics Engineering George Mason University, Virginia May 2026

GPA: 3.87

B.Tech Computer Science & Engineering JNTU Anantapur 2017

SKILLS

Languages & Frontend: Python (8+ yrs), JavaScript, TypeScript, React.js, SQL, PHP, Java, Bash — FastAPI, Django, REST APIs

AI Eval & Agentic: LangGraph, LangChain, LangSmith, Agentic Workflows, Tool Orchestration, Durable State Management, HITL, Trajectory Analysis, LLM Evaluation, LLM-as-Judge, Eval-Driven Development, Reward Modeling, RLHF, RLAIF, Golden Datasets, Data Labeling, Annotation, Regression Harnesses, Trace-Level Observability, Pass@k, Scoring Methodology, Hallucination Detection, RAG Pipelines, ChromaDB, MCP, OpenAI API, Anthropic API

Infra & Observability: AWS (S3, EC2, RDS, Lambda, SQS, Step Functions), Docker, Kubernetes, Terraform, CI/CD (GitHub Actions), OpenTelemetry, Prometheus, Grafana, Distributed Systems, Microservices, PostgreSQL, Redis, RabbitMQ, Kafka, Multi-Tenant Architecture